THE EXTENDED ALTERNATING CECH COMPLEX AS A COLIMIT OF KOSZUL COMPLEXES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Koszul complex of a finite sequence and its differential on exterior powers. constructed in the same chapter, source lines 6620-6700 and 7100-7298. Supplies the complexes whose colimit is being compared.
- Extended alternating Cech complex, defined through the localisations at products of the elements. defined in the same section, source lines 6908-6944, immediately before the counted statement. Supplies the target of the comparison.
- Colimits of complexes are computed degreewise and a map out of a colimit is a compatible system of maps on the finite pieces. used at proof lines 7120-7190. Makes the comparison maps and their compatibility meaningful.

Source reference. Original lemma 0913: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11431–11431. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0623: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7782–7782. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0621: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7780–7780. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let R be a ring. Let $f_1, \ldots, f_r \in R$. The extended alternating Čech complex*

$$R \to \bigoplus\nolimits_{i_0} R_{f_{i_0}} \to \bigoplus\nolimits_{i_0<i_1} R_{f_{i_0}f_{i_1}} \to \ldots \to R_{f_1\ldots f_r}$$

is a colimit of the Koszul complexes $K(R, f_1^n, \ldots, f_r^n)$; see proof for a precise statement.

Proof. We urge the reader to prove this for themselves. Denote $K(R, f_1^n, \ldots, f_r^n)$ the Koszul complex of Definition 0623 viewed as a cochain complex sitting in degrees $0, \ldots, r$. Thus we have

$$K(R, f_1^n, \ldots, f_r^n) : 0 \to \wedge^r(R^{\oplus r}) \to \wedge^{r-1}(R^{\oplus r}) \to \ldots \to R^{\oplus r} \to R \to 0$$

with the term $\wedge^r(R^{\oplus r})$ sitting in degree 0. Let $e_1^n, \ldots, e_r^n$ be the standard basis of $R^{\oplus r}$. Then the elements $e_{j_1}^n \wedge \ldots \wedge e_{j_{r-p}}^n$ for $1 \leq j_1 < \ldots < j_{r-p} \leq r$ form a basis for the term in degree p of the Koszul complex. Further, observe that

$$d(e_{j_1}^n \wedge \ldots \wedge e_{j_{r-p}}^n) = \sum (-1)^{a+1} f_{j_a}^n e_{j_1}^n \wedge \ldots \wedge \hat{e}_{j_a}^n \wedge \ldots \wedge e_{j_{r-p}}^n$$

by our construction of the Koszul complex in Section 0621. The transition maps of our system

$$K(R, f_1^n, \ldots, f_r^n) \to K(R, f_1^{n+1}, \ldots, f_r^{n+1})$$

are given by the rule

$$e_{j_1}^n \wedge \ldots \wedge e_{j_{r-p}}^n \longmapsto f_{i_0} \ldots f_{i_{p-1}} e_{j_1}^{n+1} \wedge \ldots \wedge e_{j_{r-p}}^{n+1}$$

where the indices $1 \leq i_0 < \ldots < i_{p-1} \leq r$ are such that $\{1, \ldots r\} = \{i_0, \ldots, i_{p-1}\} \amalg \{j_1, \ldots, j_{r-p}\}$. We omit the short computation that shows this is compatible with differentials. Observe that the transition maps are always 1 in degree 0 and equal to $f_1 \ldots f_r$ in degree r.

Denote $K^p(R, f_1^n, \ldots, f_r^n)$ the term of degree p in the Koszul complex. Observe that for any $f \in R$ we have

$$R_f = \operatorname{colim}(R \xrightarrow{f} R \xrightarrow{f} R \to \ldots)$$

Hence we see that in degree p we obtain

$$\operatorname{colim} K^p(R, f_1^n, \ldots f_r^n) = \bigoplus\nolimits_{1 \leq i_0 < \ldots < i_{p-1} \leq r} R_{f_{i_0} \ldots f_{i_{p-1}}}$$

Here the element $e_{j_1}^n \wedge \ldots \wedge e_{j_{r-p}}^n$ of the Koszul complex above maps in the colimit to the element $(f_{i_0} \ldots f_{i_{p-1}})^{-n}$ in the summand $R_{f_{i_0} \ldots f_{i_{p-1}}}$ where the indices are chosen such that $\{1, \ldots r\} = \{i_0, \ldots, i_{p-1}\} \amalg \{j_1, \ldots, j_{r-p}\}$. Thus the differential on this complex is given by

$$d(1 \text{ in } R_{f_{i_0} \ldots f_{i_{p-1}}}) = \sum\nolimits_{i \notin \{i_0, \ldots, i_{p-1}\}} (-1)^{i-t} \text{ in } R_{f_{i_0} \ldots f_{i_t} f_i f_{i_{t+1}} \ldots f_{i_{p-1}}}$$

Thus if we consider the map of complexes given in degree p by the map

$$\bigoplus\nolimits_{1 \leq i_0 < \ldots < i_{p-1} \leq r} R_{f_{i_0} \ldots f_{i_{p-1}}} \longrightarrow \bigoplus\nolimits_{1 \leq i_0 < \ldots < i_{p-1} \leq r} R_{f_{i_0} \ldots f_{i_{p-1}}}$$

determined by the rule

$$1 \text{ in } R_{f_{i_0} \ldots f_{i_{p-1}}} \longmapsto (-1)^{i_0 + \ldots + i_{p-1} + p} \text{ in } R_{f_{i_0} \ldots f_{i_{p-1}}}$$

then we get an isomorphism of complexes from $\operatorname{colim} K(R, f_1^n, \ldots, f_r^n)$ to the extended alternating Čech complex defined in this section. We omit the verification that the signs work out. $\square$